

HDOC - 14

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Subject – C programming

❖ PROBLEM STATEMENT:

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all its digits are unique.
- Find the frequency of each digit from `0` to `9`.

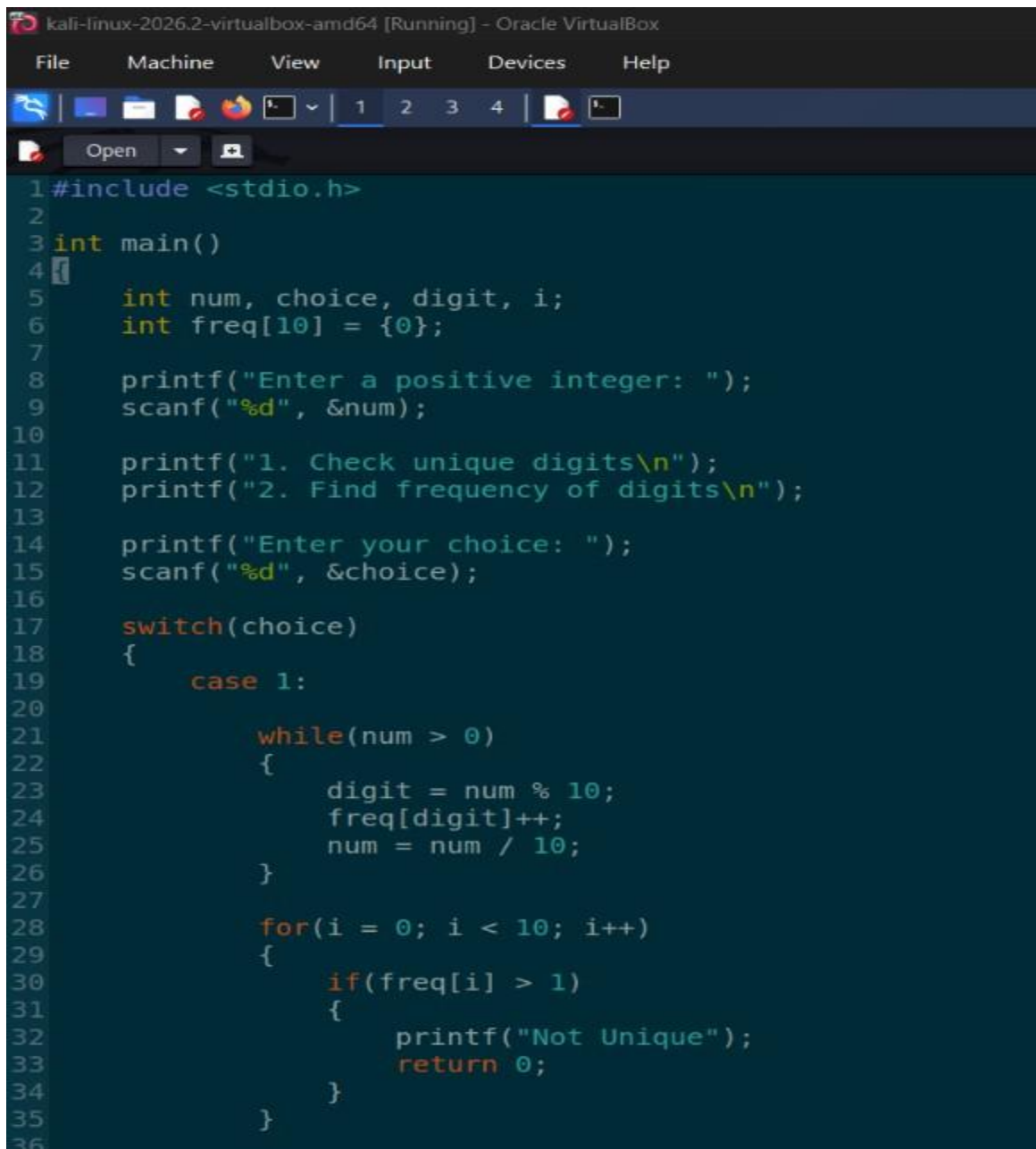
○ Algorithm:

- 1) Start.
- 2) Input a positive integer.
- 3) Display the menu:
 - 1: Check unique digits
 - 2: Find frequency of digits
- 4) Input the user's choice.
- 5) Use switch-case.
- 6) For **choice 1**, check each digit. If any digit occurs more than once, print **Not Unique**; otherwise print **Unique**.
- 7) For **choice 2**, count how many times each digit from 0 to 9 occurs and display the frequencies.
- 8) If the choice is not 1 or 2, print **Invalid Choice**.
- 9) Stop.

○ TEST CASE TABLE:

Test case	Input	choice	Expected output	Actual output	Result
1.	12345	1	Unique	Unique	Pass
2.	2389432	1	Not unique	Not unique	Pass
3.	3947130	2	Freq_0 = 1 Freq_1 = 1 Freq_2 = 0 Freq_3 = 2 Frrq_4 = 1 Freq_5 = 0 Freq_6 = 0 Freq_7 = 1 Freq_8 = 0 Frrq_9 = 1	Freq_0 = 1 Freq_1 = 1 Freq_2 = 0 Freq_3 = 2 Frrq_4 = 1 Freq_5 = 0 Freq_6 = 0 Freq_7 = 1 Freq_8 = 0 Frrq_9 = 1	Pass
4.	1213242	2	Freq_0 = 0 Freq_1 = 2 Freq_2 = 3 Freq_3 = 1 Frrq_4 = 1 Freq_5 = 0 Freq_6 = 0 Freq_7 = 0 Freq_8 = 0 Frrq_9 = 0	Freq_0 = 0 Freq_1 = 2 Freq_2 = 3 Freq_3 = 1 Frrq_4 = 1 Freq_5 = 0 Freq_6 = 0 Freq_7 = 0 Freq_8 = 0 Frrq_9 = 0	Pass
5.	3	3	Invalid choice	Invalid choice	Pass

- C Program:

A screenshot of a terminal window titled "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". Below the menu is a toolbar with icons for file operations and a tab bar showing tabs 1, 2, 3, and 4. The main area displays a C program. The code includes a header file, declares variables, prompts the user for a positive integer and a choice, and uses a switch statement to handle the choice. For choice 1, it checks if the digits of the number are unique by using an array to track frequency. If any digit appears more than once, it prints "Not Unique" and returns 0.

```
1#include <stdio.h>
2
3int main()
4{
5    int num, choice, digit, i;
6    int freq[10] = {0};
7
8    printf("Enter a positive integer: ");
9    scanf("%d", &num);
10
11    printf("1. Check unique digits\n");
12    printf("2. Find frequency of digits\n");
13
14    printf("Enter your choice: ");
15    scanf("%d", &choice);
16
17    switch(choice)
18    {
19        case 1:
20
21            while(num > 0)
22            {
23                digit = num % 10;
24                freq[digit]++;
25                num = num / 10;
26            }
27
28            for(i = 0; i < 10; i++)
29            {
30                if(freq[i] > 1)
31                {
32                    printf("Not Unique");
33                    return 0;
34                }
35            }
36
```



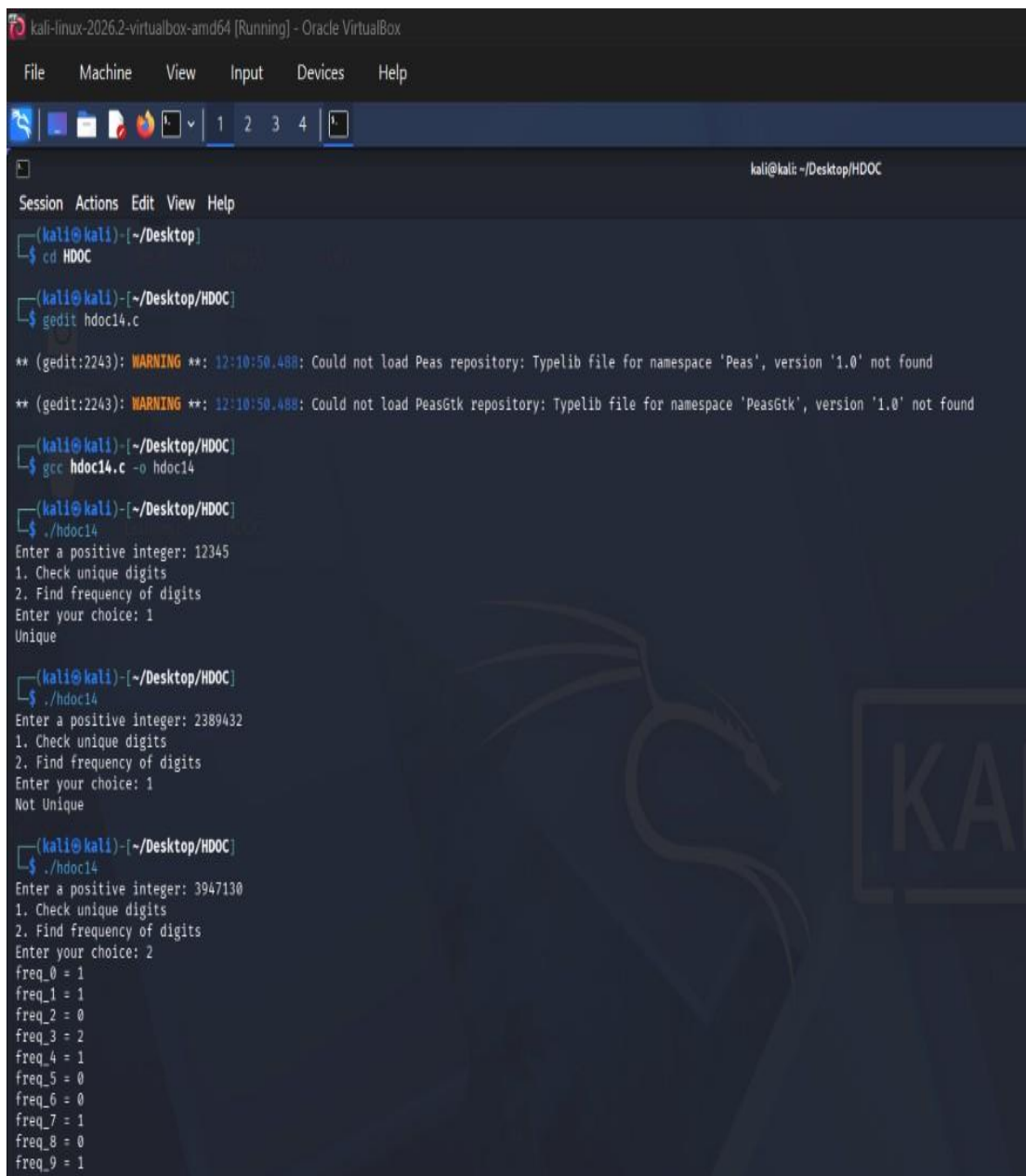
```
28     for(i = 0; i < 10; i++)
29     {
30         if(freq[i] > 1)
31         {
32             printf("Not Unique");
33             return 0;
34         }
35     }
36
37     printf("Unique");
38     break;
39
40
41     case 2:
42
43         while(num > 0)
44         {
45             digit = num % 10;
46             freq[digit]++;
47             num = num / 10;
48         }
49
50         for(i = 0; i < 10; i++)
51         {
52             printf("freq_%d = %d\n", i, freq[i]);
53         }
54
55         break;
56
57
58     default:
59
60         printf("Invalid Choice");
61     }
62
63     return 0;
64 }
```

Compile it suing:

`gcc hdoc14.c -o hdoc14`

Execute the program using:

`./hdoc14`



```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/Desktop/HDOC
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ cd HDOC
(kali@kali)-[~/Desktop/HDOC]
$ gedit hdoc14.c
** (gedit:2243): WARNING **: 12:10:50.488: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2243): WARNING **: 12:10:50.488: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
(kali@kali)-[~/Desktop/HDOC]
$ gcc hdoc14.c -o hdoc14
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc14
Enter a positive integer: 12345
1. Check unique digits
2. Find frequency of digits
Enter your choice: 1
Unique

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc14
Enter a positive integer: 2389432
1. Check unique digits
2. Find frequency of digits
Enter your choice: 1
Not Unique

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc14
Enter a positive integer: 3947130
1. Check unique digits
2. Find frequency of digits
Enter your choice: 2
freq_0 = 1
freq_1 = 1
freq_2 = 0
freq_3 = 2
freq_4 = 1
freq_5 = 0
freq_6 = 0
freq_7 = 1
freq_8 = 0
freq_9 = 1
```

```
(kali@kali)-[~/Desktop/HDOC]
└─$ ./hdoc14
Enter a positive integer: 1213242
1. Check unique digits
2. Find frequency of digits
Enter your choice: 2
freq_0 = 0
freq_1 = 2
freq_2 = 3
freq_3 = 1
freq_4 = 1
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

```
(kali@kali)-[~/Desktop/HDOC]
└─$ ./hdoc14
Enter a positive integer: 3
1. Check unique digits
2. Find frequency of digits
Enter your choice: 3
Invalid Choice
```

```
(kali@kali)-[~/Desktop/HDOC]
```